Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 8

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}
```

In [3]:
```
t1=set(map(int,input("Enter a set A:").split()))
t2=set(map(int,input("Enter a set B:").split()))

print("Union :",t1.union(t2))
print("Intersection :",t1.intersection(t2))
print("Difference(A-B) :",t1.difference(t2))
```

```
Union : {1, 2, 3, 4, 5, 6}
Intersection : {3, 4}
Difference(A-B) : {1, 2}
```

02) WAP to Find elements common to all three given sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}
```

Sample Output:

```
{2, 3}
```

In [4]:
```
t1 = set(map(int,input("Enter a set A:").split()))
t2 = set(map(int,input("Enter a set B:").split()))
t3 = set(map(int,input("Enter a set C:").split()))

print(t3.intersection(t1.intersection(t2)))
```

```
{2, 3}
```

03) WAP to remove duplicate elements from a list while preserving unique values.

Sample Input:

```
[10, 20, 10, 30, 20, 40]
```

Sample Output:

```
{10, 20, 30, 40}
```

In [12]:
```
t1 = list(map(int,input("Enter a set A:").split()))

t2 =set(t1)
print(t1)
print(t2)
```

```
[10, 20, 10, 30, 20, 40]
{40, 10, 20, 30}
```

04) WAP to count how many elements are common between two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
2
```

In [11]:
```
t1 = set(map(int,input("Enter a set A:").split()))
t2 = set(map(int,input("Enter a set B:").split()))
result = t1 & t2
print(len(result))
```

2

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

In [33]:
```
keys = list(input("Enter keys for dict :").split())
values = list(map(int,input("Enter value for dict :").split()))
dict1 = {k: v for k, v in zip(keys, values)}

keys = list(input("Enter keys for dict :").split())
values = list(map(int,input("Enter value for dict :").split()))
dict2 = {k: v for k, v in zip(keys, values)}

result = { k : dict1[k] for k in dict1.keys()}
for key, value in dict2.items():
    if key in result:
        result[key] += int(value)
    else:
        result[key] = int(value)
print(result)
```

```
{'1': 10, '2': 60, '3': 80, '4': 60}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

In [35]:
```
keys = list(input("Enter keys for dict :").split())
values = list(map(int,input("Enter value for dict :").split()))
dict1 = {k: v for k, v in zip(keys, values)}
print("original dictionary :", dict1)
dict1 = dict( (sorted(dict1.items())))
print("Sorted dictionary :", dict1)
```

```
original dictionary : {'Math': 70, 'Physics': 85, 'Chemistry': 60}
Sorted dictionary : {'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.

Sample Input:

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B
Min Value Key: C
```

In [41]:
```
keys = list(input("Enter keys for dict :").split())
values = list(map(int,input("Enter value for dict :").split()))
dict1 = {k: v for k, v in zip(keys, values)}

max_value = max(dict1.values())
max_key = [k for k, v in dict1.items() if v == max_value]
min_value = min(dict1.values())
min_key = [k for k, v in dict1.items() if v == min_value]
```

```
print(f"The maximum key is: {max_key}")
print(f"The minimum key is: {min_key}")
```

```
The maximum key is: ['B']
The minimum key is: ['C']
```

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

```
N = 5
```

Sample Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

In [43]:
```
number = int(input("Enter value of N:"))
dict1=dict()
for i in range(1,number):
    dict1[i] = (i*i)
print(dict1)
```

```
{1: 1, 2: 4, 3: 9, 4: 16}
```

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

```
programming
```

Sample Output:

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

In [46]:
```
word = input("Enter a word:")
dict1=dict()
```

```
for i in word:
    if i in dict1 :
        dict1[i] += 1
    else:
        dict1[i] = 1
print(dict1)
```

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

In [49]:
```
keys = list(input("Enter keys for dict :").split())
values = list(input("Enter value for dict :").split())
dict1 = {k: v for k, v in zip(keys, values)}

inv_di = {}
for k,v in dict1.items():
 inv_di[v] = k
print(inv_di)
```

```
{'A': '1', 'B': '2', 'C': '3'}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

```
Topper: Amit
```

In [2]:
```python
n = int(input("Number of students: "))
students = {}

for _ in range(n):
    name = input("Name: ")
    marks = list(map(int, input("Enter marks: ").split()))
    students[name] = marks

top = max(students, key=lambda x: sum(students[x]))
print("Top Student:", top)
```

```
Top Student: Amit
```

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

In [6]:
```python
keys = list(input("Enter keys for dict : ").split())
values = list(input("Enter values for dict : ").split())
dict1 = {k: int(v) for k, v in zip(keys, values)}

even_keys = list(map(int ,[k for k, v in dict1.items() if v % 2 == 0]))
print("Keys with even values:", even_keys)
```

```
Keys with even values: [2, 4]
```

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

```
[2, 3]
```

In [11]:
```python
keys1 = list(input("Enter keys for dict1 : ").split())
values1 = list(input("Enter values for dict1 : ").split())
dict1 = {int(k): v for k, v in zip(keys1, values1)}

keys2 = list(input("Enter keys for dict2 : ").split())
values2 = list(input("Enter values for dict2 : ").split())
dict2 = {int(k): v for k, v in zip(keys2, values2)}

common_keys = list(dict1.keys() & dict2.keys())
print("Common keys:", common_keys)
```

```
Common keys: [2, 3]
```

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

```
8
```

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

```
Key Not Found
```

In [15]:
```
keys = list(input("Enter keys for dict : ").split())
values = list(input("Enter values for dict : ").split())
dict1 = {k: int(v) for k, v in zip(keys, values)}

key = input("Enter key to search: ")

# if key in dict1:
#     print(dict1[key])
# else:
#     print("Key not found")

print(dict1.get(key, "Key not found"))
```

```
Key not found
```

15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

In [14]:
```
keys = list(input("Enter names : ").split())
values = list(input("Enter marks : ").split())
dict1 = {k: int(v) for k, v in zip(keys, values)}
```

```
result = {k: v for k, v in dict1.items() if v >= 75}
print("Filtered dictionary:", result)
```

```
Filtered dictionary: {'Amit': 78, 'Neha': 85}
```

In [ ]: